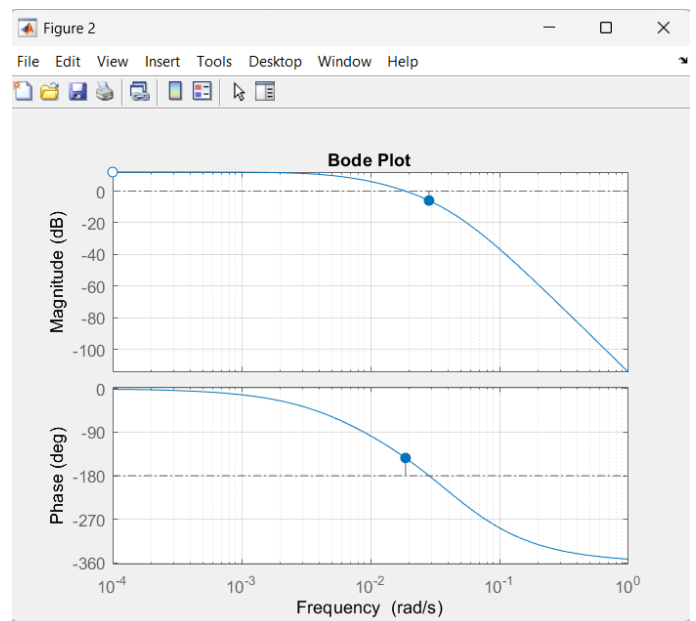
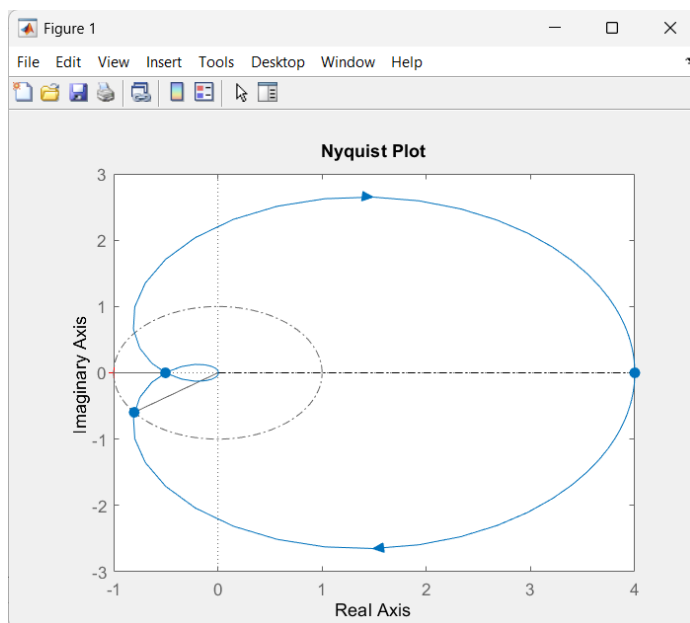


Task 1

Nyquist and Bode Plots

In this task, the open-loop transfer function $G_1(s)$ was constructed using the given parameters. The system stability was analyzed using Bode and Nyquist plots.

From the Bode plot, the magnitude decreases with increasing frequency, and both the gain margin and phase margin are positive, which indicates stable closed-loop behavior. The Nyquist plot shows that the curve does not encircle the critical point $(-1, j0)$. According to the Nyquist stability criterion, this confirms that the system is stable.

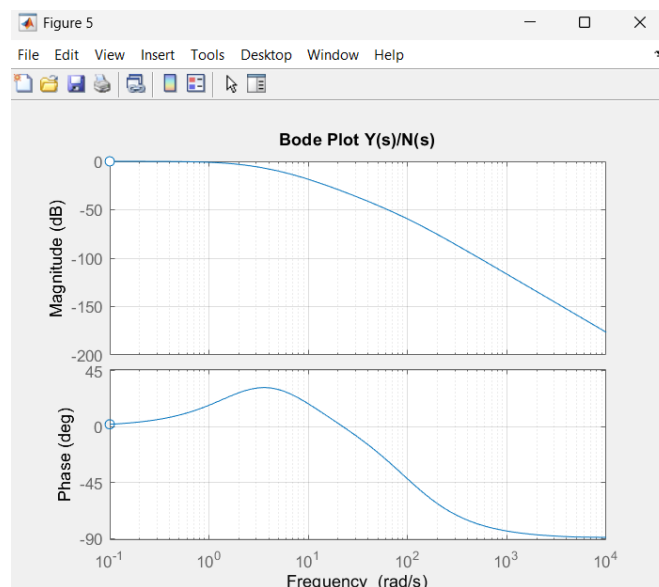
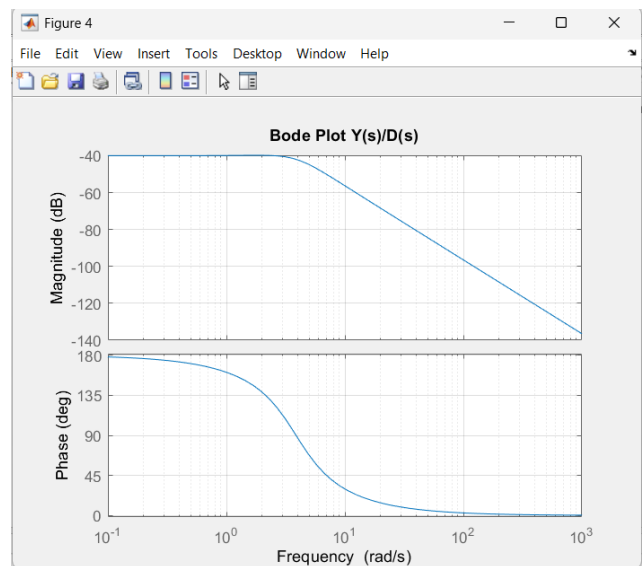
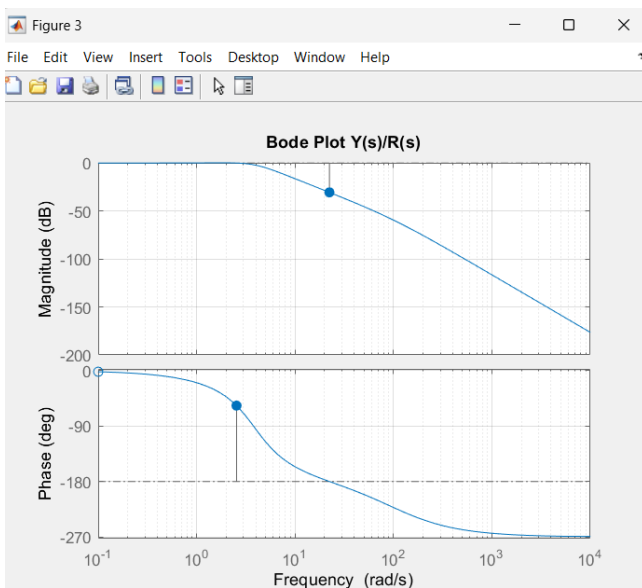
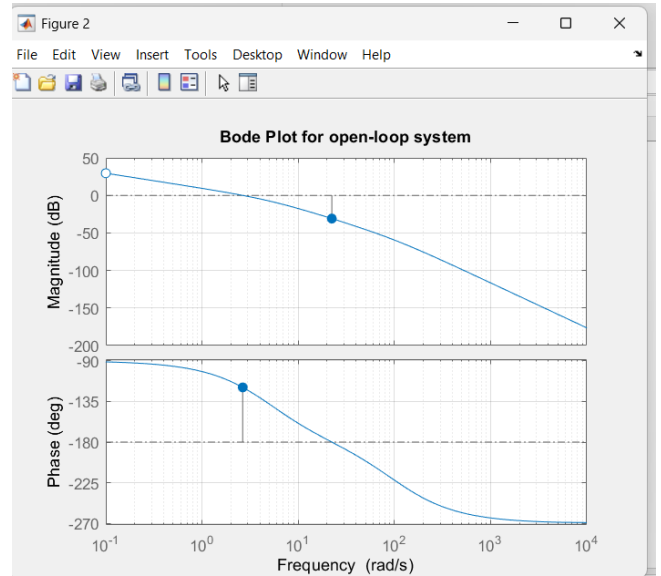
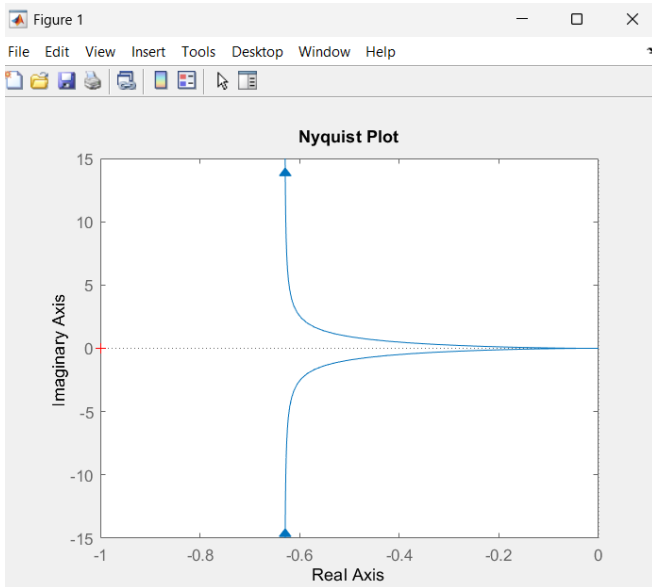


Task 2

System Stability under Different Inputs

In this task, the system responses to reference $R(s)$, disturbance $D(s)$, and noise $N(s)$ were analyzed using Bode plots.

The plots for $Y(s)/R(s)$ and $Y(s)/D(s)$ indicate stable behavior, showing that the system performs well in reference tracking and disturbance rejection. However, the response for $Y(s)/N(s)$ reveals high-frequency sensitivity, meaning the system can be affected by measurement noise.



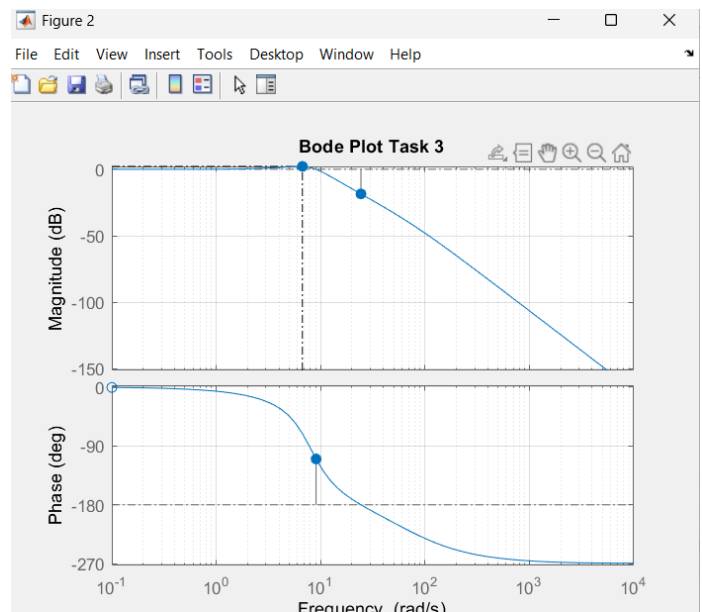
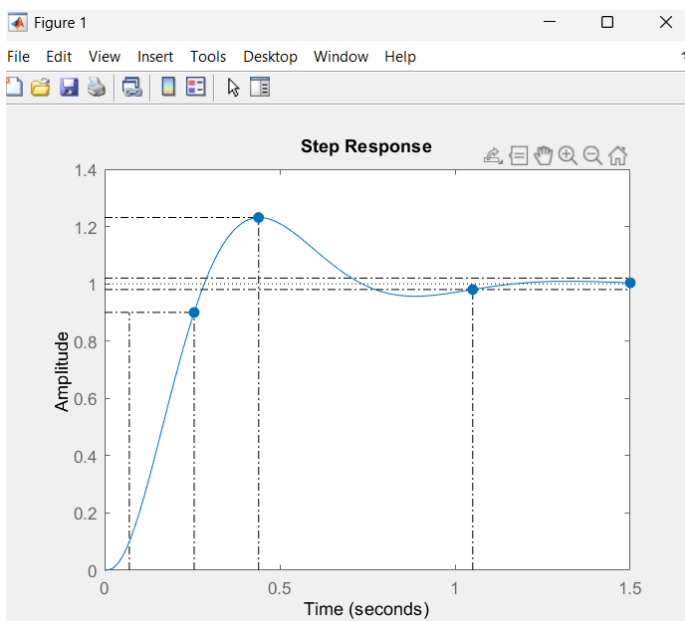
Task 3

Frequency Domain Specifications

In this task, the relationship between the frequency-domain response and the time-domain behavior of the closed-loop system was analyzed.

From the Bode plot, the resonant frequency was found to be 6.6955 rad/s with a peak magnitude of 1.2844. The presence of this resonant peak indicates that the system amplifies signals around this frequency.

The step response shows an overshoot of 23.2% and a settling time of 1.05 s. This confirms the theoretical relationship that a higher resonant peak in the frequency domain results in a larger overshoot in the time domain.



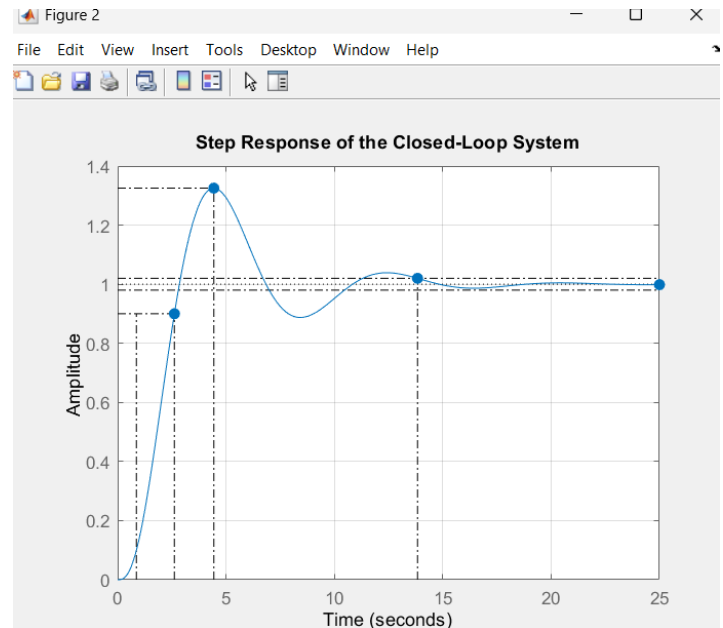
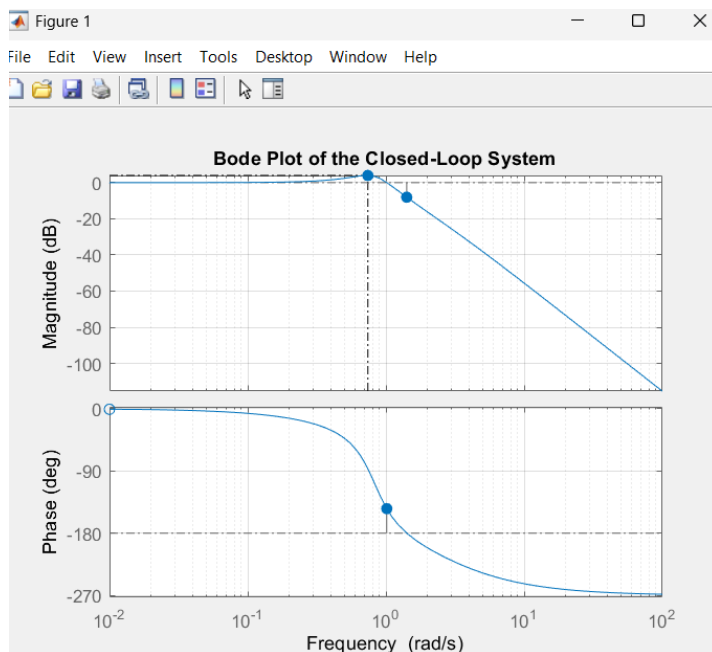
Task 4

High-Order System Analysis

In this task, a 4th-order closed-loop system was analyzed using frequency and time domain responses.

From the Bode plot, the resonant frequency was measured as 0.7399 rad/s with a peak magnitude of 3.97 dB. The relatively low resonant frequency indicates a slower system response.

The step response shows an overshoot of 32.52% and a settling time of 13.83 s, indicating a more oscillatory and slower behavior compared to the system analyzed in Task 3.



Calculation for Task 3;

$$M_{pw} = 2,17 \text{ dB} ; \omega_2 = 6,69 \text{ rad/s}$$

$$\textcircled{1} M_{pw} = 10^{(2,17/20)} = \underline{1,28}$$

$$\textcircled{2} \text{ Finding damping ratio: } M_{pw} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

$$1,28 = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \Rightarrow \boxed{\zeta \approx 0,44}$$

$$\textcircled{3} \text{ Finding ~~overshoot~~ P.O: } P.O = e^{-\left(\frac{0,44\pi}{\sqrt{1-0,44^2}}\right)} \cdot 100\% \approx \boxed{21,6\%}$$

* Matlab results: 23.2% $\rightarrow$ difference 1.6%

$$\textcircled{4} \text{ Finding } T_s : \omega_n = \frac{\omega_2}{\sqrt{1-2\zeta^2}} = \frac{6,69}{\sqrt{1-2\cdot(0,44)^2}} \approx \underline{8,49 \text{ rad/s}}$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{0,44 \cdot 8,49} = \boxed{1,07 \text{ s}}$$

* Matlab result: 1,05 s $\rightarrow$ difference 0,02 s

Task 4;

$$M_{pw} = 3,97 \text{ dB} ; \omega_2 = 0,739 \text{ rad/s}$$

$$\textcircled{1} M_{pw} = 10^{(3,97/20)} = 1,57$$

$$\textcircled{2} \text{ Finding damping ratio: } 1,57 = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \Rightarrow \boxed{\zeta \approx 0,34}$$

$$\textcircled{3} \text{ Finding P.O: } P.O = e^{-\left(\frac{0,34\pi}{\sqrt{1-0,34^2}}\right)} \cdot 100\% \approx \boxed{31,2\%}$$

* Matlab result: 32,52% $\rightarrow$ difference 1,32%

④ Finding T_s ; $\omega_n = \frac{\omega_2}{\sqrt{1 - 2\zeta^2}}$

$$\omega_n = \frac{0,739}{\sqrt{1 - 2 \cdot 0,34^2}} = \frac{0,739}{\sqrt{1 - 0,23}} = \frac{0,739}{0,87} \approx \underline{0,84 \text{ rad/s}}$$

$$T_s = \frac{4}{\zeta \omega_n} = \frac{4}{0,34 \cdot 0,84} = \frac{4}{0,28} \approx \boxed{13,94 \text{ s}}$$

Matlab result $13,83 \text{ s} \rightarrow \text{difference } \underline{0,11 \text{ s}}$